USN | | | | | | | | | | [24MTCSE102]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

STATISTICAL METHODS AND OPTIMIZATION TECHNIQUES

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Samples of size 2 are taken from the population 1, 2, 3, 4,5,6 without replacement. Determine **i.** The mean of the population. **ii.** Standard deviation of the population. **iii.** The mean of the sampling distribution of means. **iv.** The standard deviation of the sampling distribution of means.	**1**	**3**	**12**
	b.	Explain the census method and the sampling method in detail with suitable examples.	**1**	**2**	**08**
2.	**a.**	Define null and alternative hypothesis. A random sample of 10 boys had the following I.Q's: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean IQ of 100? $\left(Given\ t_{0.025,9} = 2.262\right)$	**2**	**3**	**12**
	b.	A sample of 64 students have a mean weight of 70 kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs? $\left(Given\ Z_{0.05} = 1.96\right)$	**2**	**3**	**08**
3.	**a.**	Explain the properties of good estimators with suitable examples.	**3**	**2**	**12**
	b.	In a clinical trial for a new drug, 150 patients were treated, and 30 experienced significant improvement in their condition. **i.** What is the maximum likelihood estimator (MLE) for the probability that a patient will improve after taking the drug? **ii.** If you were to treat 1,000 patients, how many would you expect to see improvement based on the MLE?	**3**	**3**	**08**
		OR			
4.	**a.**	Discuss the following with suitable examples: **i.** Point Estimation **ii.** Interval Estimation	**3**	**2**	**08**

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	b.	A teacher wants to understand the relationship between the number of hours students study and their corresponding test scores. The following data was collected from a group of students:	**3**	**3**	**12**

Hours Spend(X)	1	2	3	4	5
Test Score (Y)	50	55	65	70	80

Using the data, estimate the linear regression equation that describes how study hours affect test scores. Also, estimate the test score when students spend 6.5 hours.

5.	**a.**	Define unbounded solution and no solution stages which arise while solving LPP with graphical method. Solve the following LPP using graphical method: $Maximize\ Z = 2x_1 + 4x_2$ $Subject\ to$ $x_1 + 2x_2 \le 5$ $x_1 + x_2 \le 4$ $x_1, x_2 \ge 0$	**4**	**3**	**10**
	b.	Explain the following terminologies: **i.** Feasible solution **ii.** Feasible region **iii.** Optimal solution **iv.** Infeasible solution	**4**	**2**	**10**
		OR			
6.	**a.**	Solve the following LPP using Simplex method $Max\ Z = 6x + 8y$ $Subject\ to$ $5x + 10y \le 60$ $4x + 4y \le 40$ $x, y \ge 0$	**4**	**3**	**12**
	b.	Write R program to solve the following LPP: $Minimize\ Z = 20x + 10y$ $Subject\ to$ $x + 2y \le 40$ $3x + y \ge 30$ $4x + 3y \ge 60$ $x, y \ge 0$	**4**	**3**	**08**

7.	**a.**	Find initial basic feasible solution using Vogel's Approximation method; also find optimal solution using MODI method	**5**	**3**	**12**

	D1	D2	D3	D4	Supply
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

	b.	Write R program to solve the following Transportation problem:	**5**	**3**	**08**

Destinations					
Sources		1	2	3	Supply
	1	20	20	30	100
	2	40	10	20	150
	3	10	30	10	400
	Demand	200	150	300	65

OR

8.	**a.**	Use the Hungarian method to solve the following assignment problem, where the matrix entries represent the processing times in hours.	**6**	**3**	**12**

		Operator				
		1	2	3	4	5
Job	1	10	12	15	12	8
	2	7	16	14	14	11
	3	13	14	7	9	9
	4	12	10	11	13	10
	5	8	13	15	11	15

	b.	Write R program to solve the following Assignment problem:	**6**	**3**	**08**

Job	Machine			
	M1	M2	M3	M4
J1	5	7	11	6
J2	8	5	9	6
J3	4	7	10	7
J4	10	4	8	3